# 1.2 Measuring and Constructing Segments

Name _____ Period ____ Date _____

Geometry 1.2 · Student sheet

**I can…**

- find the length of a segment on a number line or ruler.
- tell whether two segments are congruent.
- use the Segment Addition Postulate to find lengths.

**Vocabulary** · fill in the missing word

**Postulate** — a rule that is _____ without proof

**Congruent segments** — segments that have the same _____

**Between** — B is between A and C when A, B, and C are _____ and B is on $\overline{AC}$

## Part 1 · Discovery: measure and copy

Draw $\overline{AB}$ and a point C on it. Measure in centimeters.

| AC | CB | AC + CB | AB | Length of my copy |
|---|---|---|---|---|
| | | | | |

What do you notice about AC + CB and AB?

## Part 2 · Notes and examples (fill in each note, then work the example)

**Ruler Postulate:** the distance between two points on a number line is the _____ of the difference of their coordinates: $AB = |x_2 - x_1|$.

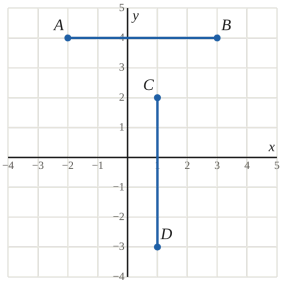
### Example 1

On a number line, A is at −3 and B is at 5. Find AB.

**Congruent segments:** $\overline{AB} \cong \overline{CD}$ means the segments have the same _____: AB = CD.

### Example 2

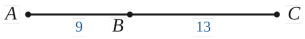
A(−2, 4), B(3, 4), C(1, 2), and D(1, −3). Is $\overline{AB} \cong \overline{CD}$?




**Segment Addition Postulate:** if B is between A and C, then AB + _____ = AC.

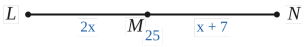
### Example 3

B is between A and C. AB = 9 and BC = 13. Find AC.




### Example 4

M is between L and N. LM = 2x, MN = x + 7, and LN = 25. Find x, LM, and MN.




### Example 5

Along a straight highway, city B is between cities A and C. It is 128 miles from A to B and 205 miles from A to C. How far is it from B to C?

## Part 3 · Partner practice

### Partner 1

On a number line, R is at −7 and S is at 4. Find RS.

### Partner 2

X is between W and Y. WX = 3x − 1, XY = 2x + 4, and WY = 28. Find x, WX, and XY.